

STUDIO 5000 ADD-ON INSTRUCTIONS

Define Add-On Instruction

1. Create and **Name** the Add-On Instruction.
2. Enter data Tags as **Parameters** to create the Add-On Instructions **Data Type**.
3. Create the **Logic** for the Add On Instruction

Use Add-On Instruction in the Program

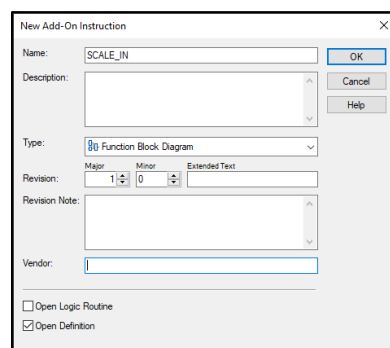
4. In the application program create a **Tag** using the Add-On Instructions **Data Type**.
5. Create the required additional application **Tags** for the Add-On Instructions **Input Data** and **Output Data**.
6. Use the Add-On Instruction into the application program and assign the Tags.

1. Creating a New Add On Instruction

To create an Add-On Instruction, right click in the Controller Organizer **Assets - Add-On Instructions**. Select the **New Add-On Instruction...** from the context menu.

Enter a **Name** and select language **Type**.

Enter any needed documentation.

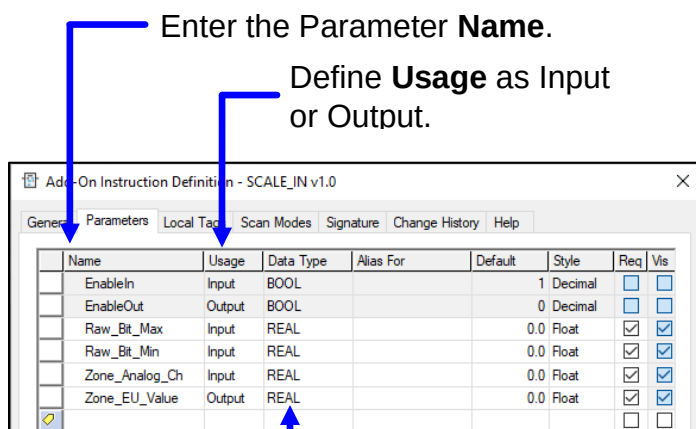


The 'New Add-On Instruction' dialog box contains the following fields and options:

- Name:** SCALE_IN
- Description:** (empty text area)
- Type:** Function Block Diagram
- Revision:** Major: 1, Minor: 0, Extended Text: (empty)
- Revision Note:** (empty text area)
- Vendor:** (empty text field)
- ☐ Open Logic Routine
- ☒ Open Definition

2. Entering the Parameters for the Add-On Instruction

See last page to how to open dialog if closed.



The 'Add-On Instruction Definition - SCALE_IN v1.0' dialog box has tabs for General, Parameters, Local Tags, Scan Modes, Signature, Change History, and Help. The Parameters tab is active, showing a table with the following data:

Name	Usage	Data Type	Alias For	Default	Style	Req	Vis
EnableIn	Input	BOOL		1 Decimal		☐	☐
EnableOut	Output	BOOL		0 Decimal		☐	☐
Raw_Bit_Max	Input	REAL		0.0 Float		☒	☒
Raw_Bit_Min	Input	REAL		0.0 Float		☒	☒
Zone_Analog_Ch	Input	REAL		0.0 Float		☒	☒
Zone_EU_Value	Output	REAL		0.0 Float		☒	☒

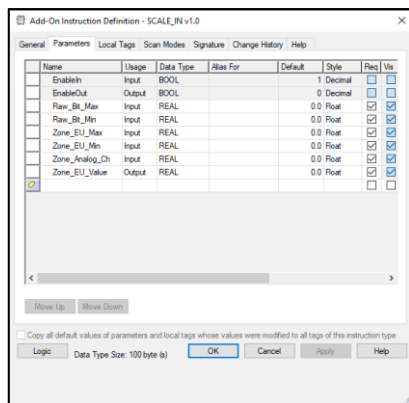
Define **Data Type**.

Enter the Parameter **Name**.

Define **Usage** as Input or Output.

Specify if **Required** and **Visible**.

Add-On Instruction Definition **Parameters** tab.



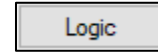
The 'Add-On Instruction Definition - SCALE_IN v1.0' dialog box has tabs for General, Parameters, Local Tags, Scan Modes, Signature, Change History, and Help. The Parameters tab is active, showing a table with the following data:

Name	Usage	Data Type	Alias For	Default	Style	Req	Vis
EnableIn	Input	BOOL		1 Decimal		☐	☐
EnableOut	Output	BOOL		0 Decimal		☐	☐
Raw_Bit_Max	Input	REAL		0.0 Float		☒	☒
Raw_Bit_Min	Input	REAL		0.0 Float		☒	☒
Zone_Analog_Ch	Input	REAL		0.0 Float		☒	☒
Zone_EU_Value	Output	REAL		0.0 Float		☒	☒

3. Entering Logic for the Add-On Instruction

See last page to how to open dialog if closed.

Select the **Logic** button at the bottom left of the Add-On Instruction Definition dialog.



Enter the program logic into the Editor as required.

The logic for the Add-On Instruction may be:

- Ladder Logic
- Function Block
- Structured Text

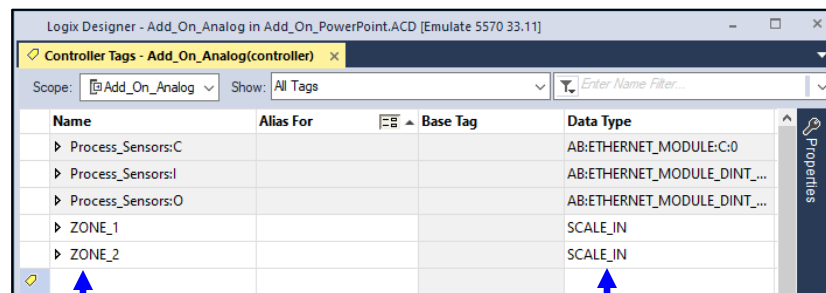
If the Add-On Instruction Definition dialog is opened, select the OK button.

4. Create Application Tags using the Add-On Data Type

Each Add-On Instruction in the application program must have a Tag defined with the Add-On Instruction's **Data Type**.

The **Data Type** name is the same as the **Add-On Instruction** name.

In this example the Add-On Instruction is named SCALE_IN.

A screenshot of the Logix Designer software interface. The title bar reads "Logix Designer - Add_On_Analog in Add_On_PowerPoint.ACD [Emulate 5570 33.11]". Below the title bar is a tab labeled "Controller Tags - Add_On_Analog(controller)". The main area contains a table with columns: "Name", "Alias For", "Base Tag", and "Data Type". The table lists several tags, including "Process_Sensors:C", "Process_Sensors:I", "Process_Sensors:O", "ZONE_1", and "ZONE_2". The "Data Type" column for "ZONE_1" and "ZONE_2" is "SCALE_IN". A blue arrow points from the text "Tag for each Add-On Instruction in the Application program." to the "Name" column. Another blue arrow points from the text "Add-On Instruction Data Type." to the "Data Type" column.

Name	Alias For	Base Tag	Data Type
▸ Process_Sensors:C			AB:ETHERNET_MODULE:C:0
▸ Process_Sensors:I			AB:ETHERNET_MODULE_DINT_...
▸ Process_Sensors:O			AB:ETHERNET_MODULE_DINT_...
▸ ZONE_1			SCALE_IN
▸ ZONE_2			SCALE_IN

Tag for each Add-On
Instruction in the
Application program.

Add-On Instruction
Data Type.

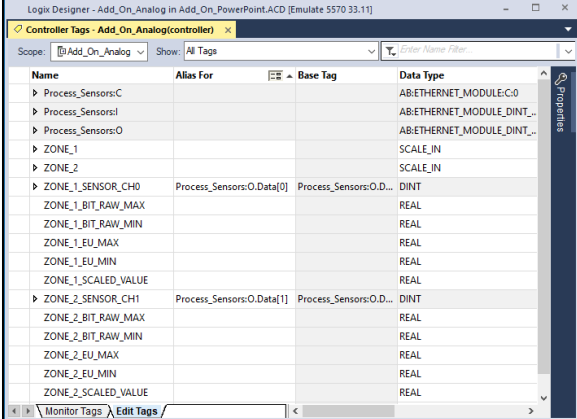
5. Create Application Tags for the Add-On Input and Output Data

This example uses two Add-On Instructions to scale data from two analog sensors, **Zone 1** and **Zone 2**

Zone 1 and Zone 2 Application Tags for the Sensor Channels, Tags for scaling values, and a Tag for Scaled result.

▶ ZONE_1_SENSOR_CH0
ZONE_1_BIT_RAW_MAX
ZONE_1_BIT_RAW_MIN
ZONE_1_EU_MAX
ZONE_1_EU_MIN
ZONE_1_SCALED_VALUE

▶ ZONE_2_SENSOR_CH1
ZONE_2_BIT_RAW_MAX
ZONE_2_BIT_RAW_MIN
ZONE_2_EU_MAX
ZONE_2_EU_MIN
ZONE_2_SCALED_VALUE



Name	Alias For	Base Tag	Data Type
Process_Sensors:C			AB:ETHERNET_MODULE:C:0
Process_Sensors:I			AB:ETHERNET_MODULE_DINT_...
Process_Sensors:O			AB:ETHERNET_MODULE_DINT_...
ZONE_1			SCALE_IN
ZONE_2			SCALE_IN
ZONE_1_SENSOR_CH0	Process_Sensors:O.Data[0]	Process_Sensors:O.D...	DINT
ZONE_1_BIT_RAW_MAX			REAL
ZONE_1_BIT_RAW_MIN			REAL
ZONE_1_EU_MAX			REAL
ZONE_1_EU_MIN			REAL
ZONE_1_SCALED_VALUE			REAL
ZONE_2_SENSOR_CH1	Process_Sensors:O.Data[1]	Process_Sensors:O.D...	DINT
ZONE_2_BIT_RAW_MAX			REAL
ZONE_2_BIT_RAW_MIN			REAL
ZONE_2_EU_MAX			REAL
ZONE_2_EU_MIN			REAL
ZONE_2_SCALED_VALUE			REAL

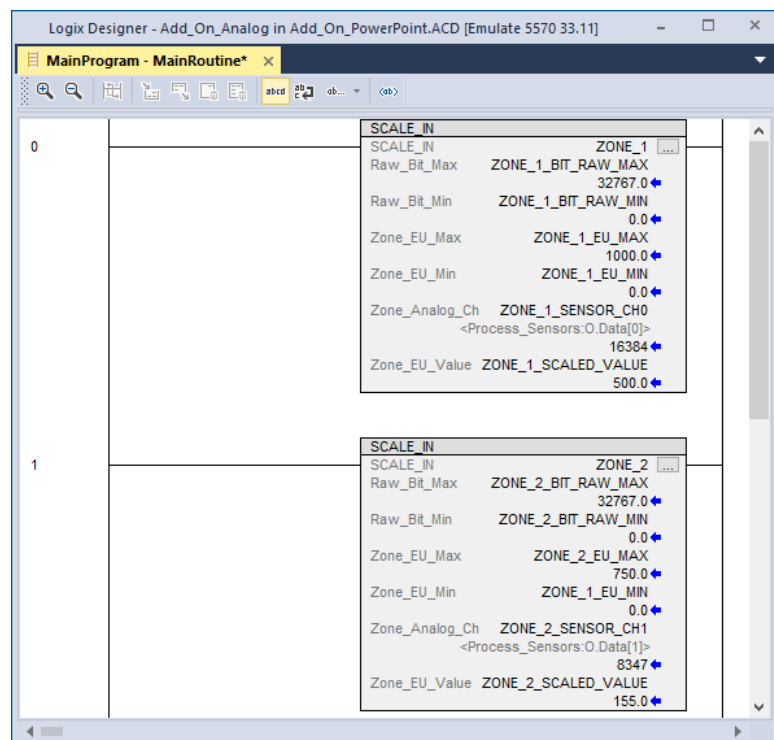
6. Use Add-On Instruction the Application Program and Assign the Tags

Drag into the Editor the Add-On Instruction.

Assign the Tag defined with the **Add-On Instruction Data Type**.

In this example **ZONE_1** and **ZONE_2** with **SCALE_IN** as the **Data Type**.

Notice the Application Tags above are used in the Add-On Instructions for **Zone 1** and **Zone 2**.



Edit Add-On Instruction After Creating

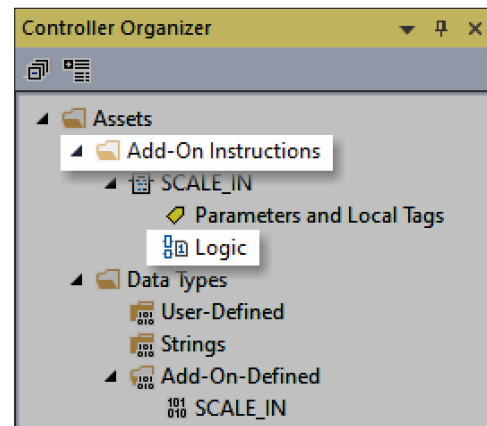
TO EDIT ADD-ON INSTRUCTION LOGIC

Expand **Assets** and **Add-On Instructions**.

Expand the named Add-On instruction.

Double left click on **Logic** to open the editor.

Logic Editor opens.



TO EDIT ADD-ON INSTRUCTION DEFINITION PARAMETERS AND LOCAL TAGS

Expand **Data Types**.

Expand **Add-On_Defined**.

Double left click on the named Add-On to open the dialog.

